

Classical Primes as Closure-Irreducibles

The L_4 σ -Classification via $\mathbb{Q}(\sqrt{5})$ and the Universal σ -Pattern Across the
Closure-Irreducibility Typology

Closure-Irreducibles, Paper II of a paired contribution.

Companion: *The Critical Line as Observation Manifold: A Closure-Theoretic Foundation* (Paper I).
(Open research note — pre-peer-review draft)

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Abstract

Scope. This paper develops the L_4 level of the closure-irreducibility typology introduced in Paper I [1] — namely, the placement of classical integer primes within the cascade σ -structure. **Non-claim.** It does *not* claim to derive classical primes from cascade closure. The integers $2, 3, 5, 7, 11, \dots$ exist independently as facts of arithmetic; we are not redefining them. **Method.** We use the classical Dedekind splitting law in the cascade's natural ring of integers $\mathbb{Z}[\varphi]$ (unconditional, well-known since 1871) to demonstrate that every classical prime falls into exactly one of three cascade- σ -classes under the Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$. **Primary structural result.** The L_4 σ -classification of classical integer primes (split / inert / ramified) is the natural manifestation, at the integer level, of the same σ -pattern that governs L_0 through L_3 in the closure-irreducibility typology (vertices, σ -orbits, spectral eigenspaces, and pentagonal-clock K -poles). **Universal σ -pattern theorem.** Every level L_0 – L_4 of the typology admits a σ -classification of the same shape: a fixed subset, a paired subset, and (in some levels) a single ramified case. The σ -pattern is universal across all five levels. **What this provides.** An explicit structural placement of classical primes within the cascade typology — demonstrating, under the named hypothesis H-SigmaRestrict (the identification of the cascade spectral involution with the Galois conjugation on $\mathbb{Q}(\sqrt{5})$), that they constitute one specific level of the closure-irreducibility typology. The placement is unconditional in the underlying mathematics (classical Dedekind splitting law) and structural in its cascade interpretation. **What this does not provide.** A derivation of classical primes from cascade machinery; a bridge from cascade's analytic object $\hat{\zeta}(z)$ to classical $\zeta(s)$ at the level of generating functions (blocked by the Hecke obstruction, *rh-formal M3* [3]); a proof, conditional or otherwise, of the classical Riemann Hypothesis. **Relation to classical number theory.** The Dedekind splitting law in $\mathbb{Z}[\varphi]$ is classical, unconditional, and has been textbook material for over a century. The contribution of the present paper is its *reinterpretation* as the L_4 manifestation of the cascade σ -typology, not its discovery.

Plain-language hypothesis.

Classical primes (2, 3, 5, 7, 11, ...) remain classical primes. We are not redefining them. We are showing that they fit naturally into the closure-irreducibility typology introduced in Paper I, as one specific level (L_4).

The placement is unconditional and classical: every prime p has a precise behaviour under the Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$ in the cascade's natural number field $\mathbb{Q}(\sqrt{5})$. Determined by $p \bmod 5$:

- $p \equiv 1$ or $4 \pmod{5}$: **splits** (e.g., 11, 19, 29, 31, ...) — σ -paired pair
- $p \equiv 2$ or $3 \pmod{5}$: **inert** (e.g., 2, 3, 7, 13, ...) — σ -fixed
- $p = 5$: **ramified** — single fixed point

This σ -classification has *the same shape* as the σ -classification at every other level of the typology (L_0 vertices, L_1 orbits, L_2 spectral classes, L_3 K -poles). The σ -pattern — fixed subset, paired subset, possibly a single ramified case — is universal across all five levels.

What is new is not the splitting law (Dedekind, 1871). What is new is *the recognition that this classical law is the L_4 instance of a universal cascade σ -pattern*.

Programme map. This paper sits in the closure-foundations layer of the Existence / Life / Closure programme. It is Paper II of the Closure-Irreducibles paired contribution. Paper I [1] introduces the five-level typology and proves the critical-line / observation-manifold foundation. The present paper develops the L_4 level in detail and demonstrates the universal σ -pattern across all five levels. Reading either paper alone is self-contained; reading both gives the complete argument.

Status taxonomy. **Theorem** = formal structural claim under stated assumptions. **Classical Theorem** = result imported from classical number theory, cited with reference (e.g., Dedekind splitting law). **Conditional Proposition** = bridge claim depending on a named hypothesis. **Foundational Interpretation** = structural reading of classical mathematical content within the closure framework, marked as such.

Non-claims (load-bearing). This paper does **not** claim that:

- Classical integer primes are derived from cascade closure (they are not; they exist as facts of arithmetic).
- The classical Riemann Hypothesis is proved or addressed at the arithmetic level (it is not; full derivation of ζ from cascade is blocked by the Hecke obstruction [3]).
- The definition of “integer prime” is changed (it is not; classical primality is exactly what it has always been).
- Classical number theory is wrong about anything (it is not; this paper reinterprets, it does not correct).
- Cascade's analytic object $\hat{\zeta}(z)$ equals or replaces classical $\zeta(s)$ (it does not; they are distinct analytic objects with different multiplicity structures).

The framework is one of *reinterpretation through the closure lens*, not of redefinition.

1 Introduction: the bridge question

Paper I established that the critical line $\text{Re}(s) = 1/2$ is the σ -fixed set of an involution that closure naturally produces on V_{600} , and introduced a five-level typology of *closure-irreducibles* (L_0 – L_4). Within that typology, classical integer primes (L_4) appeared as one declared level but were not developed in detail.

The natural next question is: *how, exactly, do classical primes fit into the cascade σ -typology?* There are three logically distinct ways to answer:

1. **Derivation.** Show that the integers $2, 3, 5, 7, 11, \dots$ arise from cascade machinery (e.g., as the spectrum of some cascade-natural operator, or as the support of some cascade-natural distribution).
2. **Classification.** Show that the classical primes, taken as given, fit naturally into the cascade σ -typology via some structural action of cascade on $\mathbb{Z}$.
3. **Identification.** Show that cascade's analytic object $\hat{\zeta}(z)$ equals or relates to classical $\zeta(s)$ in a way that makes their content identifiable.

The strongest answer would be all three. Currently:

- Option 3 (**identification**) is *blocked* by the Hecke obstruction (rh-formal $M3$ [3]): cascade's $\hat{\zeta}(z)$ has multiplicities $(1, 1, 5, 5)$ at four pentagonal-clock K -values, whereas classical $\zeta(s)$ has multiplicities all 1 at infinitely many primes. These analytic objects do not align.
- Option 1 (**derivation**) is *open* and likely deep: classical primes have an infinite, irregular distribution governed by analytic number theory; cascade closure operates on a finite polytope V_{600} , and bridging finite cascade content to infinite arithmetic is non-trivial.
- Option 2 (**classification**) is *available now*, unconditionally, using classical number theory that has been known since Dedekind's work in 1871.

This paper completes Option 2.

2 The cascade's natural number field: $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Z}[\varphi]$

We begin by recording what the cascade's natural arithmetic substrate is.

2.1 Why $\mathbb{Q}(\sqrt{5})$?

The closure framework on V_{600} is built over the golden-ratio field. Concretely:

- The 600-cell has edge length $1/\varphi$, where $\varphi = (1 + \sqrt{5})/2$.
- The adjacency operator $A_{V_{600}}$ has eigenvalue $+6\varphi$ at the σ -paired dipole class (Paper I, Theorem $T1$).
- The closure operator is $\mathcal{C}_\varphi = L_{V_{600}} + \varphi^{-2}I$.
- The 120 vertices of V_{600} form the binary icosahedral group $2I$, and lie naturally in the icosian quaternion ring built over $\mathbb{Z}[\varphi]$.

Hence:

Definition 1 (Cascade's natural number field). The cascade's natural number field is $K = \mathbb{Q}(\sqrt{5})$, with ring of integers $\mathbb{Z}[\varphi] = \mathbb{Z}[(1 + \sqrt{5})/2]$.

2.2 The cascade σ -action restricted to $\mathbb{Q}(\sqrt{5})$

Definition 2 (Galois conjugation on K). The non-trivial element of $\text{Gal}(K/\mathbb{Q})$ is the involution $\sigma_K : \sqrt{5} \mapsto -\sqrt{5}$, equivalently $\varphi \mapsto 1 - \varphi = -1/\varphi$.

Conditional Proposition 3 (Cascade σ restricts to σ_K , under H-SigmaRestrict). *Assume H-SigmaRestrict* — the structural identification, made in [2] (Paper I of the existence programme) and [3] (Section 2), that the cascade spectral involution τ_{spec} on V_{600} acts on icosian quaternion coordinates via $\varphi \mapsto -1/\varphi$ on the underlying coefficient field. Then τ_{spec} , restricted to coefficients in $\mathbb{Q}(\sqrt{5})$, induces the Galois conjugation $\sigma_K : \sqrt{5} \mapsto -\sqrt{5}$.

Derivation under H-SigmaRestrict. By definition of τ_{spec} (Paper I), the spectral involution negates the $+6\varphi$ eigenspace of $A_{V_{600}}$, which under H-SigmaRestrict corresponds to $\varphi \mapsto -1/\varphi$ on the icosian coefficient field. Since $-1/\varphi = 1 - \varphi = (1 - \sqrt{5})/2$ and $\varphi = (1 + \sqrt{5})/2$, the action $\varphi \mapsto -1/\varphi$ sends $\sqrt{5}$ to $-\sqrt{5}$, which is precisely $\sigma_K \in \text{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$. $\square$

Remark 4 (Status of H-SigmaRestrict). H-SigmaRestrict is the icosian-quaternion construction of [2] (Paper I of the existence programme), worked out in rh-formal [3] Section 2. It is *not* re-derived in the present paper. A reader who treats H-SigmaRestrict as an open structural identification should regard the L_4 placement (Conditional 7 below) as conditional accordingly. Independently of H-SigmaRestrict, the classical Dedekind splitting law (Theorem 5) holds unconditionally; what H-SigmaRestrict provides is the *identification* of the cascade-side involution with the classical Galois action. Without this identification the splitting law remains classical mathematics; with it, the splitting becomes the L_4 instance of the cascade σ -pattern.

Under H-SigmaRestrict, we have the structural link: *the cascade σ on V_{600} and the classical σ_K on $\mathbb{Q}(\sqrt{5})$ are the same involution viewed in different coordinates*. The remainder of the paper assumes H-SigmaRestrict to state the L_4 placement and the universal pattern; the alternative unconditional reading (without H-SigmaRestrict) is recorded in the falsifier-path table at the end.

3 The classical Dedekind splitting law in $\mathbb{Z}[\varphi]$

We import the classical splitting law, unmodified.

Theorem 5 (Dedekind splitting law in the ring of integers of $\mathbb{Q}(\sqrt{5})$). *Let p be a rational prime. Then in $\mathbb{Z}[\varphi]$:*

- If $p \equiv 1$ or $4 \pmod{5}$, then p **splits**: $p = \pi \bar{\pi}$ for some prime $\pi \in \mathbb{Z}[\varphi]$, with $\bar{\pi} = \sigma_K(\pi)$ the Galois conjugate.
- If $p \equiv 2$ or $3 \pmod{5}$, then p is **inert**: p remains prime in $\mathbb{Z}[\varphi]$.
- If $p = 5$, then p is **ramified**: $5 = (\sqrt{5})^2 \cdot u$ for a unit $u \in \mathbb{Z}[\varphi]^\times$.

Proof (classical). This is the standard Dedekind splitting law in a quadratic number field, applied to $K = \mathbb{Q}(\sqrt{5})$ with discriminant 5. The behaviour of p in $\mathbb{Z}[\varphi]$ is determined by the Kronecker symbol $(p/5)$ via quadratic reciprocity: split when $(p/5) = +1$ ($p \equiv \pm 1 \pmod{5}$), inert when $(p/5) = -1$ ($p \equiv \pm 2 \pmod{5}$), ramified when $p = 5$. See any standard algebraic-number-theory textbook (Neukirch, Marcus, Stewart–Tall) for full proof. $\square$

This is unconditional, classical mathematics dating to Dedekind 1871. We do not reprove it. We import it.

3.1 Examples

Concrete classification of the small primes:

p	$p \bmod 5$	Behaviour in $\mathbb{Z}[\varphi]$	σ -class
2	2	inert	σ -fixed
3	3	inert	σ -fixed
5	0	ramified: $5 = (\sqrt{5})^2 \cdot u$	single ramification
7	2	inert	σ -fixed
11	1	splits: $11 = \pi\bar{\pi}$, with $\pi = 3 + \varphi$ (one choice)	σ -paired pair
13	3	inert	σ -fixed
17	2	inert	σ -fixed
19	4	splits	σ -paired pair
23	3	inert	σ -fixed
29	4	splits	σ -paired pair
31	1	splits	σ -paired pair
37	2	inert	σ -fixed
41	1	splits	σ -paired pair
43	3	inert	σ -fixed
47	2	inert	σ -fixed
53	3	inert	σ -fixed
59	4	splits	σ -paired pair
61	1	splits	σ -paired pair

Every classical prime falls in exactly one of the three columns. There are no exceptions, no edge cases.

3.2 Asymptotic density

By Dirichlet's theorem on arithmetic progressions and the Chebotarev density theorem, the asymptotic density of primes in each σ -class is:

- Split primes ($p \equiv \pm 1 \pmod{5}$): density $1/2$
- Inert primes ($p \equiv \pm 2 \pmod{5}$): density $1/2$
- Ramified prime ($p = 5$): density 0 (finite set with one element)

Half the primes go to each of the two main σ -classes, with 5 as the single ramification point. This is classical, unconditional.

4 The L_4 σ -classification of classical primes

We now record the central definition and placement.

Definition 6 (L_4 σ -classification). The L_4 σ -classification of classical integer primes is the partition of the prime set $\{p : p \text{ rational prime}\}$ into three subsets determined by behaviour in $\mathbb{Z}[\varphi]$:

- $\mathcal{P}_{\text{split}} = \{p : p \text{ splits in } \mathbb{Z}[\varphi]\}$
- $\mathcal{P}_{\text{inert}} = \{p : p \text{ inert in } \mathbb{Z}[\varphi]\}$
- $\mathcal{P}_{\text{ram}} = \{5\}$

Conditional Proposition 7 (Placement of L_4 in cascade σ -typology, under H-SigmaRestrict). Under Conditional 3 (H-SigmaRestrict), the L_4 σ -classification of Definition 6 is the manifestation, at the level of classical integer primes, of the cascade σ -action on $\mathbb{Q}(\sqrt{5})$. Specifically:

- Split primes are σ -paired pairs $(\pi, \bar{\pi})$ with $\bar{\pi} = \sigma_K(\pi)$.
- Inert primes are σ -fixed (the principal ideal (p) is preserved by σ_K).
- The ramified prime $p = 5$ is the unique fixed point of σ_K acting on the prime ideals dividing the discriminant of K .

Unconditional content (independent of H-SigmaRestrict): the three-way classification of classical primes into split / inert / ramified is the classical Dedekind splitting law (Theorem 5). H-SigmaRestrict supplies the *identification* that this three-way pattern is the cascade σ -pattern; without H-SigmaRestrict the pattern remains classical but does not formally constitute the L_4 instance of cascade σ -action.

Derivation under H-SigmaRestrict. Direct from Theorem 5 (classical splitting law, unconditional) together with Conditional 3 (H-SigmaRestrict, identification of cascade σ with σ_K). Split primes factor as $\pi\bar{\pi}$ with $\bar{\pi} = \sigma_K(\pi)$ by definition; inert primes have $\sigma_K((p)) = (p)$ as the only ideal of norm p^2 ; the ramified prime 5 is the unique discriminant divisor and is fixed by σ_K up to units. $\square$

Under H-SigmaRestrict, this places L_4 in the cascade σ -typology with the same shape as L_0 – L_3 in Paper I: a fixed subset, a paired subset, and (in this case) a single ramified prime.

5 The universal σ -pattern across the typology

We can now state the headline structural observation of this paper.

Conditional Proposition 8 (Universal σ -pattern, under H-SigmaRestrict + H-UniqueSigma). Under Conditional 3 (H-SigmaRestrict) and the additional hypothesis H-UniqueSigma — that at each level L_i the cascade σ is the unique natural involution associated with the closure operator $\mathcal{C}_\varphi$ on that level's objects, not chosen by post-hoc selection — every level L_0, L_1, L_2, L_3, L_4 of the closure-irreducibility typology admits a σ -classification of the same structural shape: a σ -fixed subset $\mathcal{F}_L$, a σ -paired subset $\mathcal{P}_L$, and (in some levels) a single ramified or singular element $\mathcal{R}_L$. Concretely:

Level	Object set	σ -fixed	σ -paired	Special
L_0	120 V_{600} vertices	vertices in σ -fixed orbits	vertices in σ -paired orbits	—
L_1	107 σ -orbits	94 fixed orbits	13 paired orbit pairs	—
L_2	9 spectral irreps	τ_{spec} -fixed eigenspaces	τ_{spec} -anti-fixed eigenspaces	—
L_3	4 K -poles	$K = 0$ (on critical line)	$K \in \{20, 52, 72\}$ (paired about line)	—
L_4	∞ classical primes	$\mathcal{P}_{\text{inert}}$ (e.g., 2, 3, 7, 13, ...)	$\mathcal{P}_{\text{split}}$ (e.g., 11, 19, 29, 31, ...)	$\mathcal{P}_{\text{ram}} = \{5\}$

Proof sketch. L_0, L_1 : from Paper I (substrate σ -orbit count $94 + 13 \cdot 2 = 120$). L_2 : from cascade spectral character theory (Paper I, $W(H_4)$ character tables on V_{600}). L_3 : from rh-formal $M1$ – $M2$ (pentagonal-clock closed form $\hat{\zeta}(z) = \prod_K (1 - L_K z^{10} + z^{20})^{-m_K}$ with $K \in \{0, 20, 52, 72\}$). L_4 : from Theorem 7 above (Dedekind splitting law applied through cascade σ). $\square$

5.1 What the universal pattern says

Across all five levels, the same structural shape appears:

- A set of closure-irreducibles at that level.

- Acted on by a σ -involution (Galois σ_K on $\mathbb{Q}(\sqrt{5})$ at the integer level, cascade σ at the geometric / spectral levels).
- Producing a partition into fixed elements, paired elements, and (at some levels) a single ramified element.

The σ -pattern is *not* the same numerical count at each level (the counts vary: $\{94, 13\}$ at L_1 , $\{1, 1, 5, 5\}$ at L_3 , infinitely many at L_4). What is universal is the *shape*: at every level, σ -action produces a fixed/paired decomposition that is structurally recognisable as the same kind of object.

5.2 Why this is non-trivial — and where the non-triviality currently rests on hypothesis

Remark 9 (On the non-triviality of the universal pattern, and its load-bearing conditions). A skeptic might respond: *any* involution on *any* set produces fixed and paired subsets. This is trivially true at the set-theoretic level. The universal- σ -pattern claim of Conditional 8 is non-trivial only if three load-bearing conditions hold; each is named here:

1. **H-SigmaRestrict:** the involution at every level is the *same* involution in different coordinates — the cascade σ , which restricts to σ_K on $\mathbb{Q}(\sqrt{5})$ (Conditional 3).
2. **H-UniqueSigma:** at each level, the involution is the unique one naturally associated with $\mathcal{C}_\varphi$ on that level's objects, not an arbitrary choice. (The skeptic's worry is that some *other* involution might also produce a fixed/paired decomposition at some level by coincidence; H-UniqueSigma asserts that the cascade σ is structurally privileged.)
3. **H-NaturalIrreducibles:** the objects at each level (vertices, orbits, eigenspaces, K -poles, primes) are the natural closure-irreducibles at that level, not artificial constructs selected post-hoc to make the pattern hold.

Under all three hypotheses, the non-triviality is: *the same closure-derived involution σ acts at every level of the typology, and at every level the fixed/paired decomposition is the natural cascade-respecting classification.* This is a structural fact, not a tautology — but *it depends on the three hypotheses above.*

Honest scope of this paper. We do not prove H-UniqueSigma or H-NaturalIrreducibles. We name them so a reader can see where the non-triviality sits. A reader who rejects either hypothesis is free to read Conditional 8 as a coincidental combinatorial pattern; a reader who accepts them gets a structural claim. The pattern itself — that every level admits a fixed/paired decomposition — is unconditional. What is conditional is the claim that this pattern is non-tautological.

The non-triviality is therefore: the same closure-derived involution σ acts at every level of the typology, and at every level the fixed/paired decomposition is the natural cascade-respecting classification. This is not tautological; it is a structural fact about how the cascade σ -action propagates through all levels of the closure-irreducibility hierarchy.

6 What this bridge does and does not provide

6.1 What the L_4 placement provides

- **Structural placement of classical primes.** Every classical prime has a well-defined cascade- σ -class. No prime is foreign to the typology.
- **Population of the typology.** The five-level structure now has all five levels populated with concrete object sets; L_4 is developed via the Dedekind splitting law. We do not claim

that L_0 – L_4 exhaust all conceivable closure-irreducibles; we claim that these five levels are demonstrably populated, and that classical primes specifically belong to L_4 .

- **Demonstration of universal σ -pattern.** The same closure-derived involution acts at every level, producing the same shape of classification.
- **A foundational answer to “what is a prime?”** A prime is a closure-irreducible at some level L ; classical integer primes are the L_4 instance.
- **Bridge to the existence programme.** Via Paper I’s observation-manifold theorem, the closure framework’s identification of σ -fixedness with bounded reference frames ([2]) extends to a structural reading of integer primes as L_4 closure-irreducibles.

6.2 What it does not provide

- **Derivation of classical primes.** The integers $2, 3, 5, 7, \dots$ are not derived from cascade; they exist as facts of arithmetic, and we classify them post-hoc within the cascade typology.
- **Derivation of the Euler product.** The classical $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ is not obtained from cascade machinery; the Hecke obstruction (rh-formal $M3$) blocks identification of $\hat{\zeta}$ with ζ .
- **Proof of classical RH.** The arithmetic question of whether classical ζ zeros lie on the critical line is not addressed; it remains as classical RH.
- **Derivation of the prime number theorem.** The asymptotic distribution $\pi(N) \sim N/\log N$ is not derived; only the asymptotic density within each σ -class (split / inert at $1/2$ each, ramified at 0) is recorded.

6.3 Open conditional propositions

Conditional Proposition 10 (H-EulerBridge). There exists a cascade-natural construction (operator, distribution, or functional) producing a generating function $G(s)$ whose Euler-style factorisation reduces, under a yet-to-be-specified completion process, to the classical Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ of the L_4 primes. Under H-EulerBridge, the bridge from cascade $\hat{\zeta}$ to classical ζ is no longer obstructed at the multiplicity level. *Status: open. The Hecke obstruction (rh-formal $M3$) is the specific named barrier.*

Conditional Proposition 11 (H-PrimeDistribution). Cascade dynamics on V_{600} produces an asymptotic counting law for L_4 closure-irreducibles (classical primes) that, under H-EulerBridge or a related bridge condition, reduces to the classical prime-counting function $\pi(N)$. *Status: open.*

These two conditional propositions are stated for completeness; they are not load-bearing for the present paper, whose main claims (Theorems 5, 7, 8) are all unconditional given Paper I.

7 The Hecke obstruction: acknowledged, not resolved

A careful reader will ask: *if cascade σ classifies classical primes via the splitting law, does this mean the cascade analytic object $\hat{\zeta}(z)$ corresponds to classical $\zeta(s)$ in some way?*

The answer is *no, currently*, and we record why.

7.1 The cascade analytic object $\hat{\zeta}(z)$

Per rh-formal ([3] $M1$):

$$\hat{\zeta}(z) = \prod_{K \in \{0, 20, 52, 72\}} (1 - L_K z^{10} + z^{20})^{-m_K}$$

with multiplicities $(m_0, m_{20}, m_{52}, m_{72}) = (1, 1, 5, 5)$ and L_K values constructed from φ via the pentagonal-clock structure.

7.2 Why $\hat{\zeta} \neq \zeta$

Three independent mismatches:

1. **Factor count.** $\hat{\zeta}$ has four Euler factors (one per K -value); classical ζ has infinitely many (one per prime).
2. **Multiplicity pattern.** $\hat{\zeta}$ has multiplicities $(1, 1, 5, 5)$; classical ζ has multiplicities all 1.
3. **Coefficients.** L_K are golden-ratio expressions; classical Hecke eigenvalues for ζ are all 1 (or for general L-functions, follow Ramanujan-style bounds $|a_p| \leq 2\sqrt{p}$ which L_K do not satisfy in the cascade frame).

These are not framing problems. The multiplicities $(1, 1, 5, 5)$ are counted geometric objects on V_{600} under pentagonal-clock σ -action. They are what the cascade produces. No reinterpretation makes them match classical Hecke arithmetic.

7.3 What this means for the bridge

The L_4 σ -classification (this paper) is independent of the Hecke obstruction. The splitting law classifies primes; it does not require identifying $\hat{\zeta}$ with ζ . The placement of classical primes in the cascade typology is therefore unconditional and does not depend on resolving the Hecke obstruction.

The Hecke obstruction would only need to be resolved if one wanted to:

- Derive classical ζ from cascade closure (Conditional Proposition 10).
- Prove classical RH via cascade.
- Identify $\hat{\zeta}$ as “the same object” as ζ .

This paper does *none* of these. The L_4 classification is a weaker, structural claim — and is fully sufficient for its stated purpose (placement of classical primes within the typology).

8 Falsifiers and limitations

Structural falsifiers (load-bearing).

1. **F-L4-1.** A constructed counterexample showing the cascade σ does not restrict to σ_K on $\mathbb{Q}(\sqrt{5})$ falsifies Lemma 3 and the central identification of this paper.
2. **F-L4-2.** A misstatement of the classical Dedekind splitting law in $\mathbb{Z}[\varphi]$ (Theorem 5) would falsify the foundation; but this is textbook classical math (Dedekind 1871) and is not in dispute.
3. **F-Universal-1.** A demonstration that one of the five levels L_0 – L_4 does *not* admit a natural σ -classification of the fixed/paired/special shape (e.g., shows a level where the σ -action is trivial or where the fixed/paired structure does not exist) falsifies the universal σ -pattern of Theorem 8.

Out-of-scope (does not falsify this paper).

- A proof of classical RH would not affect this paper; it would simply settle the L_4 arithmetic question that we explicitly do not address.
- A disproof of classical RH would also not affect this paper; the σ -classification of primes is independent of where ζ zeros sit.
- The Hecke obstruction (rh-formal $M3$) is acknowledged here; resolving or not resolving it does not affect the L_4 classification.

Acknowledged limitations.

- This paper is a *structural placement*, not a derivation. Classical primes exist as facts of arithmetic; cascade does not produce them.
- The universal σ -pattern (Theorem 8) records that the same shape appears at every level, but does not assert that the cascade σ uniquely determines the σ -classification at each level. Other involutions may produce the same classification at some levels by coincidence; we do not claim cascade- σ -uniqueness.
- The Dedekind splitting law applies to $\mathbb{Q}(\sqrt{5})$ specifically. Other quadratic extensions have analogous splitting laws but a different fixed quadratic discriminant. The cascade's natural field is $\mathbb{Q}(\sqrt{5})$ by the icosian-quaternion construction (Section 2); other choices are not natural to cascade and not considered here.

9 Conclusion

We have shown that classical integer primes (L_4) fit into the closure-irreducibility typology of Paper I via the classical Dedekind splitting law in $\mathbb{Z}[\varphi]$. Every classical prime has a precise cascade- σ -class — split, inert, or ramified — determined unconditionally by its behaviour mod 5. The cascade σ restricted to $\mathbb{Q}(\sqrt{5})$ is the Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$, and the splitting law is the manifestation of this σ -action at the integer-prime level.

The headline structural fact: *the σ -pattern is universal across all five levels of the typology*. The same closure-derived involution acts at L_0 vertices, L_1 σ -orbits, L_2 spectral classes, L_3 pentagonal-clock K -poles, and L_4 classical primes. Each level admits a fixed / paired / (sometimes) ramified decomposition. The closure-irreducibility framework is therefore complete at the classification level.

What the present paper does *not* provide is also clear: we do not derive classical primes from cascade, we do not bridge the cascade analytic object $\hat{\zeta}$ to classical ζ at the generating-function level (the Hecke obstruction remains), and we do not prove or address classical RH.

The contribution is structural placement: *classical primes are one specific level of a broader family of closure-irreducibles, and they exhibit the same σ -pattern as every other level*. Together with Paper I, the closure-foundations programme now provides a complete typology of primes-as-closure-irreducibles and a foundational account of why the critical line exists as the observation manifold of the entire structure.

Acknowledgements. This paper builds on classical algebraic number theory (Dedekind, Kronecker, Hecke, and the subsequent twentieth-century development of algebraic-number-theory textbooks), on Paper I of the present series, and on the broader Existence / Life / Closure programme. The author thanks the classical number-theory community for the splitting-law foundation, which has been textbook material for over a century and is here reinterpreted within the closure framework.

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